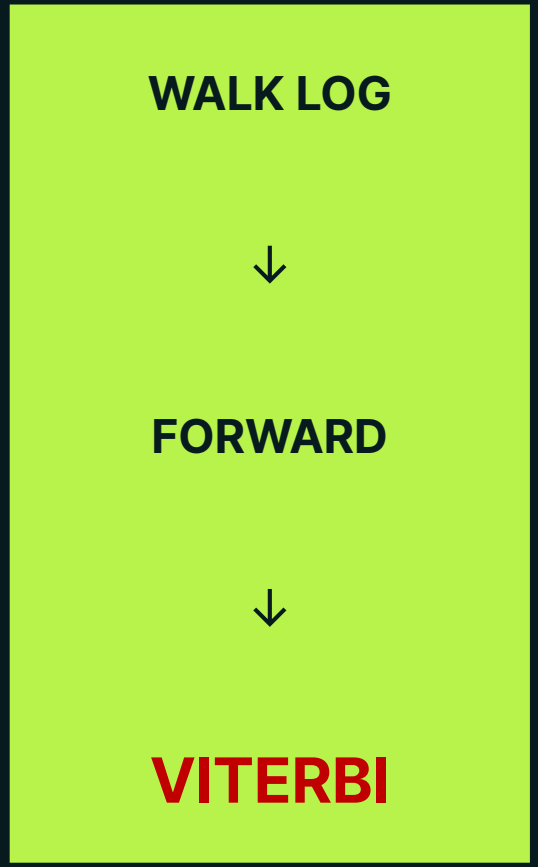
The Professor's Walk Model Forward and Viterbi by Hand

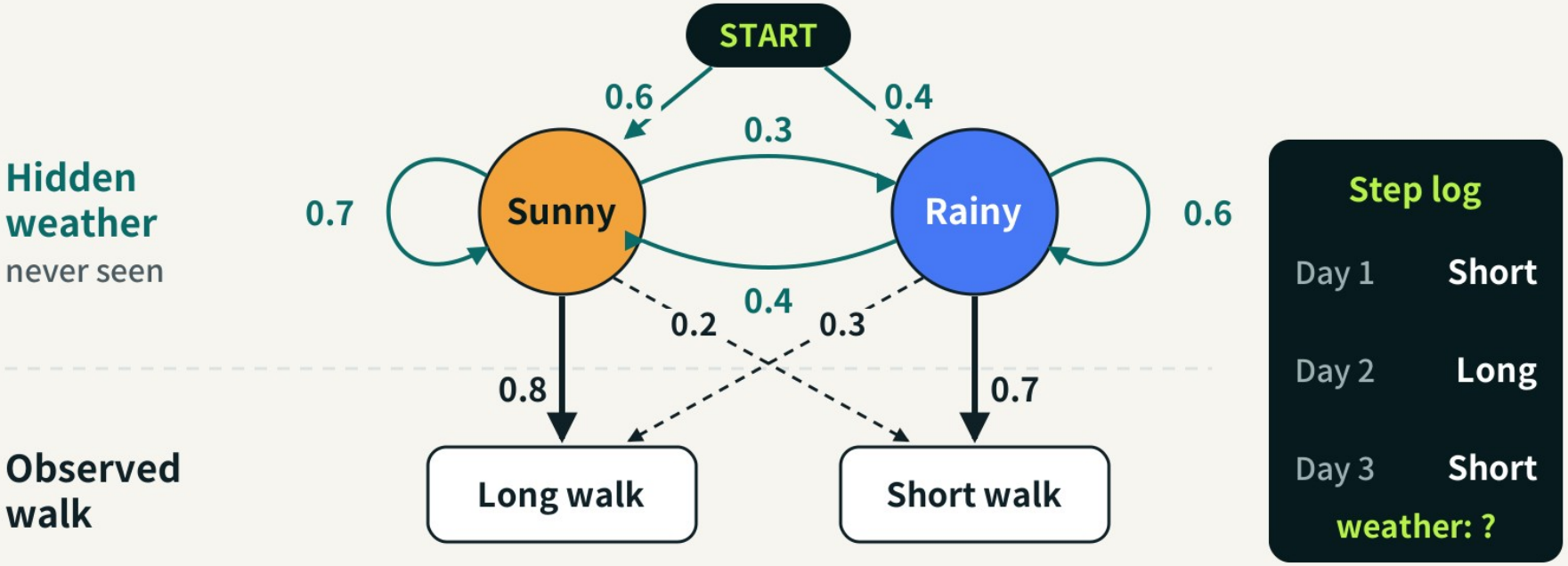
Guessing the hidden weather from three days of walks



Hidden Markov Models · Worked Example

The professor's walk depends on weather we never see

Hidden: the weather (Sunny or Rainy). Seen: whether the walk was Short or Long



※ Teal arrows: how weather changes overnight. Dark arrows: how weather sets the walk length

Three small tables hold every chance in the model

Every number in this deck comes from these three tables

How Day 1 starts
before any walk

Sunny	0.6
Rainy	0.4

Overnight change
today (row) → tomorrow (column)

	Sunny	Rainy
Sunny	0.7	0.3
Rainy	0.4	0.6

Walk given weather
weather (row) → walk (column)

	Short	Long
Sunny	0.2	0.8
Rainy	0.7	0.3

each row adds up to 1

※ Read the middle table as: if today is Sunny, tomorrow is Sunny with chance 0.7 and Rainy with chance 0.3

The walk log raises two different questions

Walk log for three days: Short, Long, Short

QUESTION 1

How likely is the log?

- Counts every weather story
- Adds them all up

Answered by the Forward algorithm



QUESTION 2

Which story is best?

- Looks for one single story
- Keeps only the best

Answered by the Viterbi algorithm

Both questions use the same three tables and the same picture.

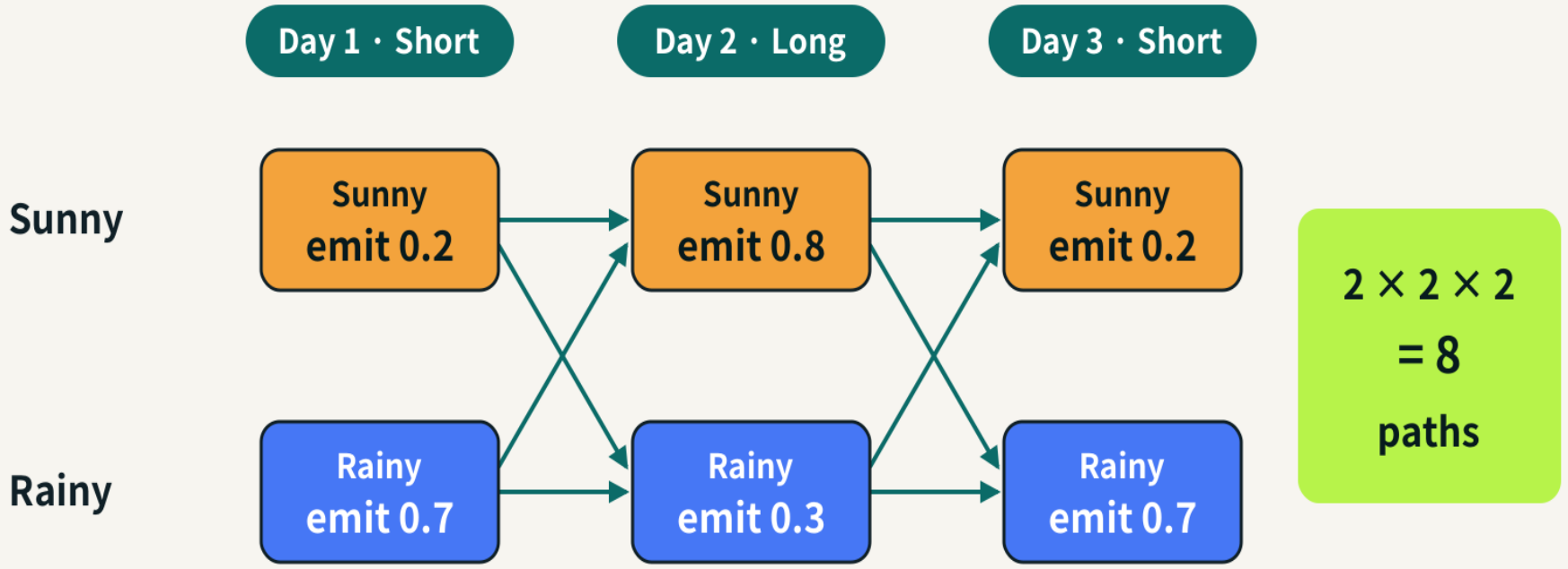
01

Listing Every Weather Story

The slow way: try all weather sequences one by one

Eight weather stories could explain the same three walks

Two kinds of weather over three days give 2 × 2 × 2 possible stories



※ Each box shows the chance that this weather produces that day's walk

One story's chance is a chain of multiplications

Example story: Sunny on all three days

$$\text{Chance of one story} = \text{start} \times \text{walk} \times \text{overnight change} \times \text{walk} \times \text{overnight change} \times \text{walk}$$



The slow way repeats this chain for all eight stories, then adds them up.

Adding all eight stories gives 0.08548

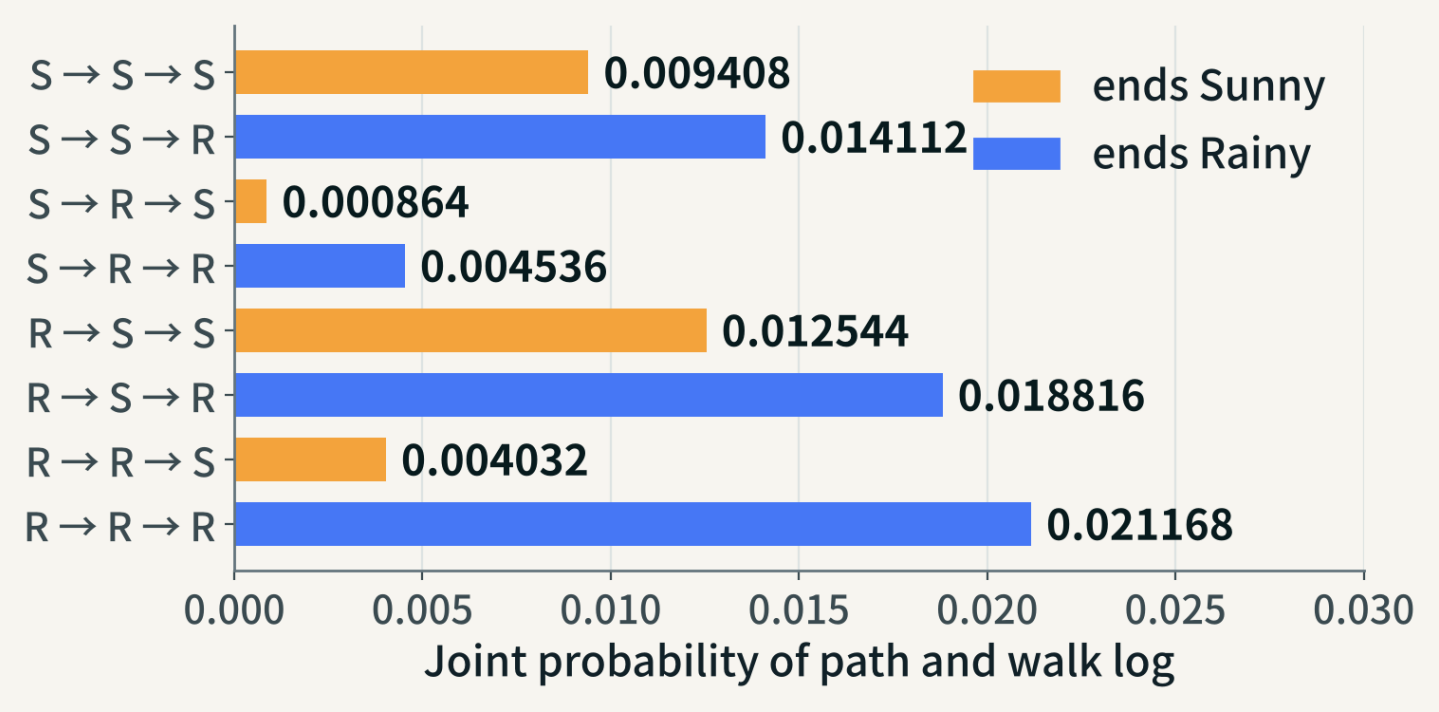
Chance of each weather story together with the walks Short, Long, Short

LIST EVERY STORY

Total 0.08548

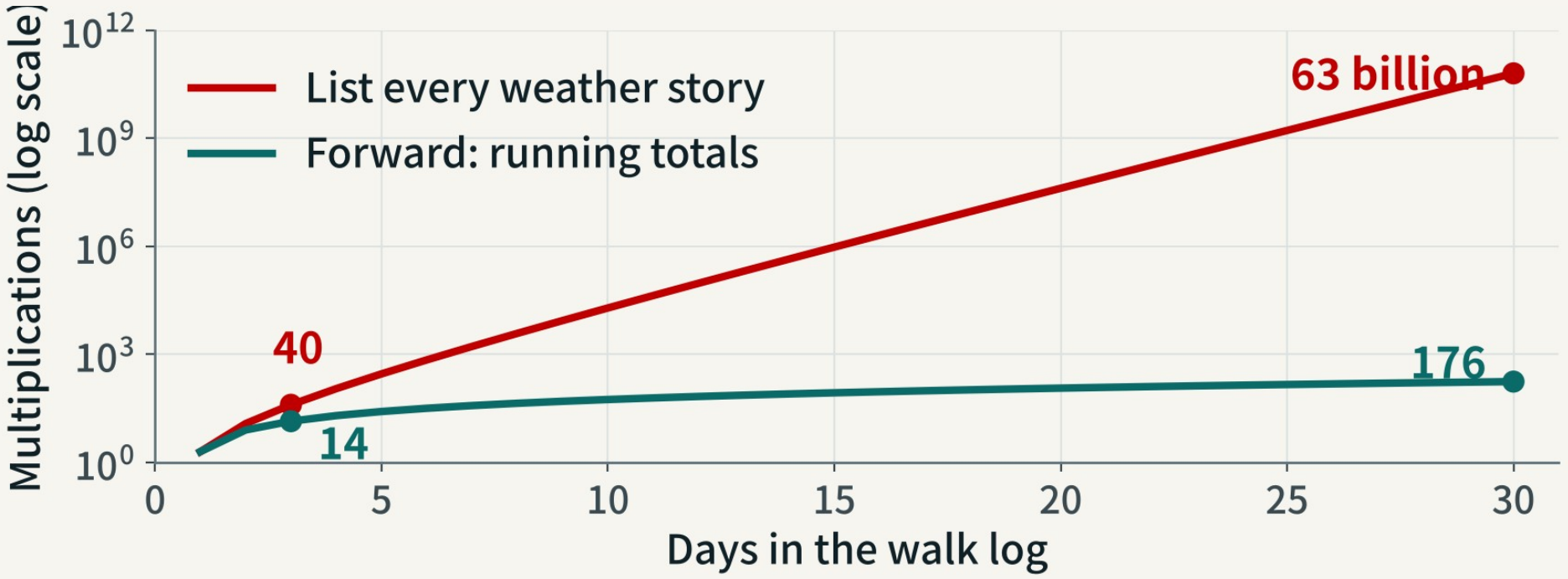
- S = Sunny, R = Rainy
- 40 multiplications

The same first steps are redone



Listing stories explodes as the walk log gets longer

Multiplications needed with two kinds of weather



※ For 30 days there are about 1.07 billion stories; running totals need only 176 multiplications

02

The Forward Algorithm

Keep one running total per weather, day by day

Each day, update two running totals with one rule

Shown for Sunny; Rainy works the same way with its own numbers

The forward rule in words

$$\text{Sunny total today} = \big(\text{Sunny total yesterday} \times \text{stays sunny} + \text{Rainy total yesterday} \times \text{turns sunny}\big) \times \text{today's walk if sunny}$$

STEP 1

Bring in yesterday

one total per weather

STEP 2

Apply the night

stay or switch weather

STEP 3

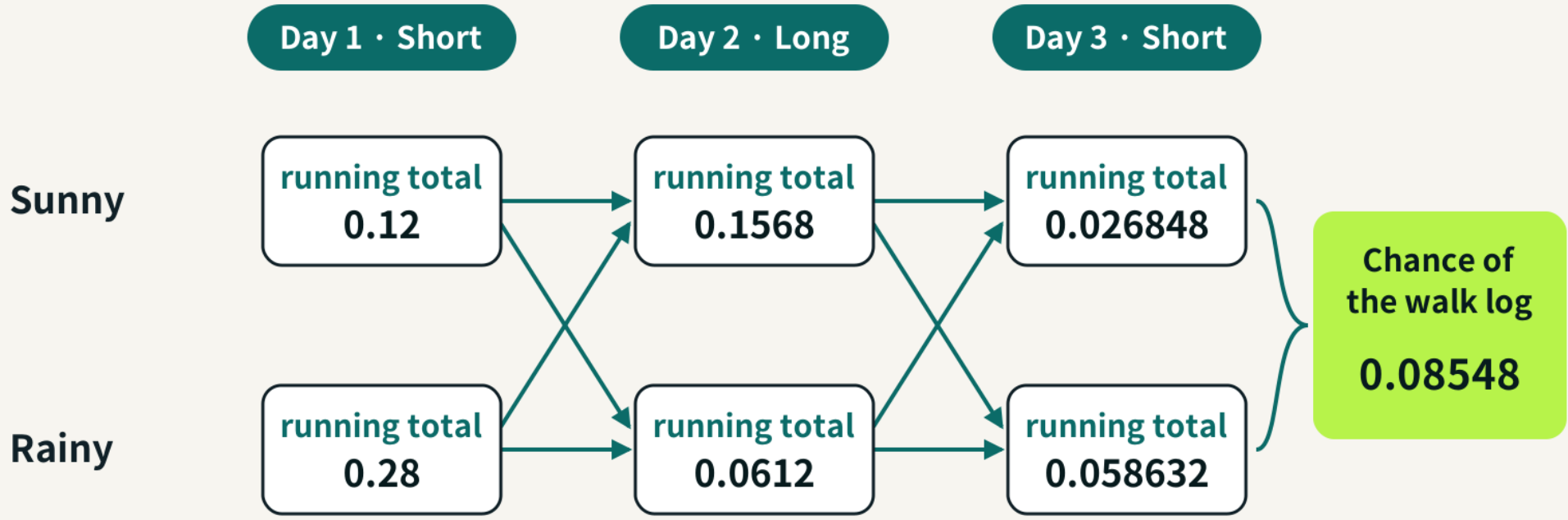
Match today's walk

times the walk chance

Yesterday's two totals are all the history we need.

Two running totals per day replace eight stories

Running totals for the walks Short, Long, Short



※ Arrows bring in yesterday's totals; each box then multiplies by that day's walk chance

Day 1: start chance times the chance of a Short walk

No yesterday yet, so each total is just two numbers multiplied

Sunny on Day 1 · walk: Short

start	0.6 chance Day 1 is sunny	
walk	× 0.2 Short walk if sunny	= 0.12

Sunny total, Day 1 = 0.12

Rainy on Day 1 · walk: Short

start	0.4 chance Day 1 is rainy	
walk	× 0.7 Short walk if rainy	= 0.28

Rainy total, Day 1 = 0.28

0.12 and 0.28 are reused by every later step.

Day 2: add both routes in, then multiply by the walk chance

Walk on Day 2 was Long

Sunny on Day 2 · walk: Long

from Sunny	0.12 × 0.7 yesterday × stays sunny	= 0.084
from Rainy	0.28 × 0.4 yesterday × turns sunny	= 0.112
add	0.084 + 0.112 both routes together	= 0.196
walk	× 0.8 Long walk if sunny	= 0.1568

Sunny total, Day 2 = 0.1568

Rainy on Day 2 · walk: Long

from Sunny	0.12 × 0.3 yesterday × turns rainy	= 0.036
from Rainy	0.28 × 0.6 yesterday × stays rainy	= 0.168
add	0.036 + 0.168 both routes together	= 0.204
walk	× 0.3 Long walk if rainy	= 0.0612

Rainy total, Day 2 = 0.0612

Each Day 1 total is used twice, not once per story.

Day 3: the two final totals add up to 0.08548

Walk on Day 3 was Short

Sunny on Day 3 · walk: Short		
from Sunny	0.1568 × 0.7 yesterday × stays sunny	= 0.10976
from Rainy	0.0612 × 0.4 yesterday × turns sunny	= 0.02448
add	0.10976 + 0.02448 both routes together	= 0.13424
walk	× 0.2 Short walk if sunny	= 0.026848
Sunny total, Day 3 = 0.026848		

Rainy on Day 3 · walk: Short		
from Sunny	0.1568 × 0.3 yesterday × turns rainy	= 0.04704
from Rainy	0.0612 × 0.6 yesterday × stays rainy	= 0.03672
add	0.04704 + 0.03672 both routes together	= 0.08376
walk	× 0.7 Short walk if rainy	= 0.058632
Rainy total, Day 3 = 0.058632		

Adding 0.026848 and 0.058632 gives 0.08548, the same as the slow way.

The running totals equal the eight stories grouped by Day 3 weather

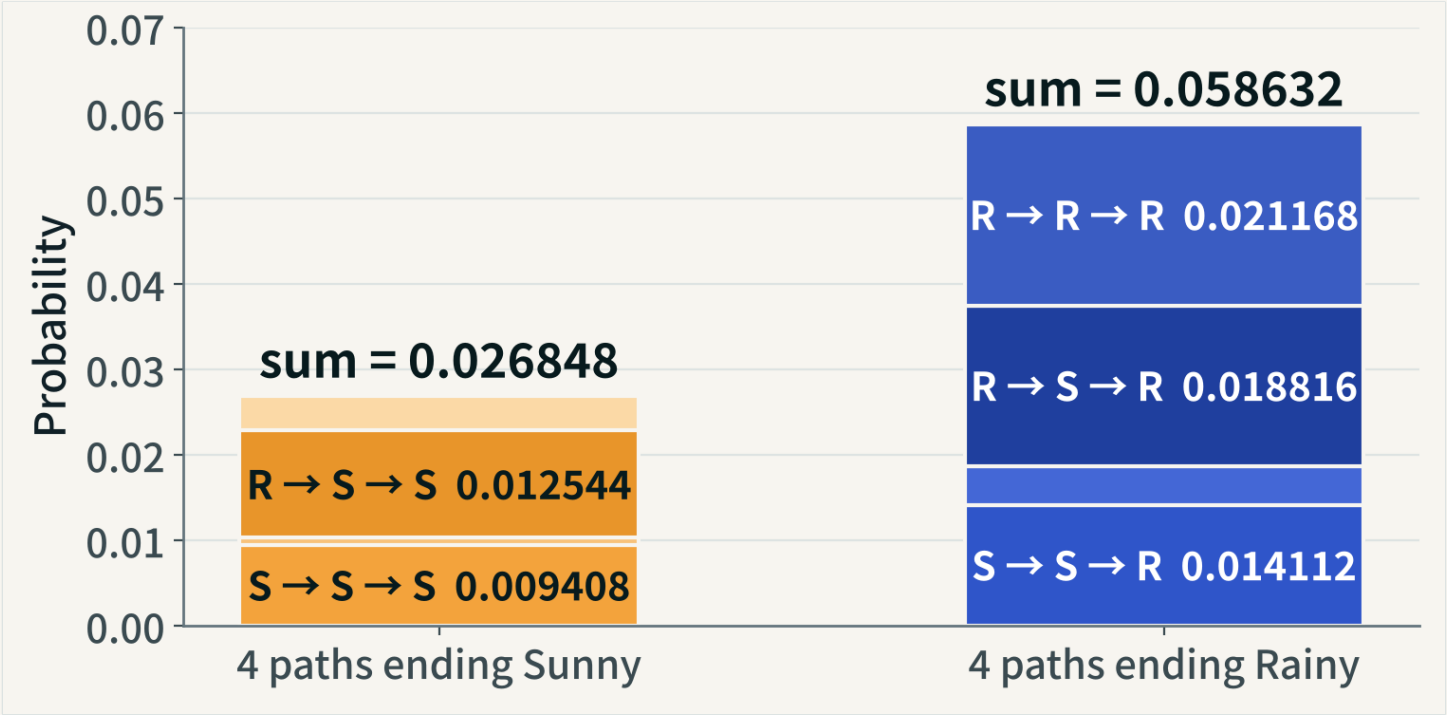
The eight story chances, stacked by the weather on the last day

WHY IT MATCHES

Same sum

- Sunny ends: 0.026848
- Rainy ends: 0.058632

Common steps done once



03

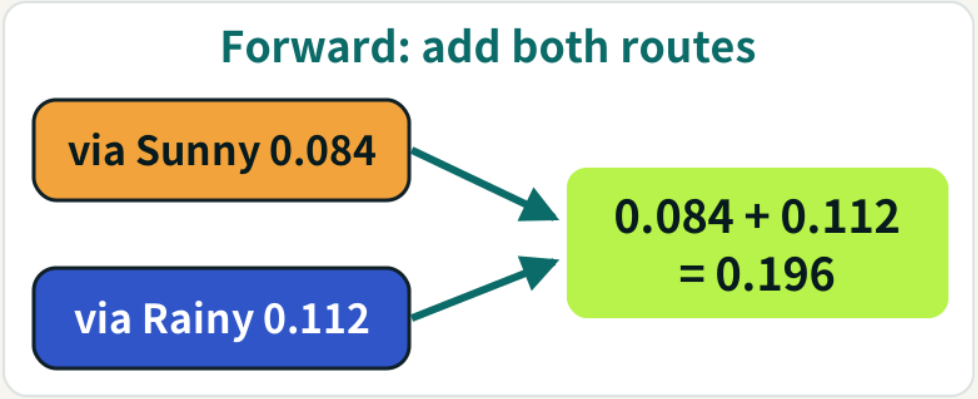
The Viterbi Algorithm

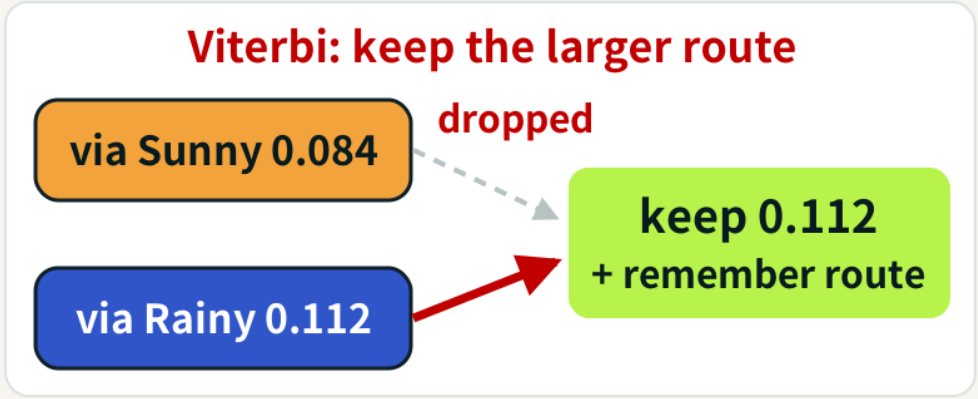
Find the single most likely weather story

Viterbi changes one word in the rule: add becomes keep the larger

The Viterbi rule in words

$$\text{Sunny best today} = \text{larger of}\big(\text{Sunny best yesterday} \times \text{stays sunny},\ \text{Rainy best yesterday} \times \text{turns sunny}\big) \times \text{today's walk if sunny}$$





Each box keeps its best score and remembers which route won.

Day 2: each box keeps only its stronger incoming route

Day 1 scores are the same as the forward start: 0.12 and 0.28

Best way into Sunny, Day 2		
from Sunny	0.12 × 0.7 best × stays sunny	= 0.084 dropped
from Rainy	0.28 × 0.4 best × turns sunny	= 0.112
keep	0.112 route from Rainy wins	larger
walk	× 0.8 Long walk if sunny	= 0.0896
0.0896 · Rainy → Sunny		

Best way into Rainy, Day 2		
from Sunny	0.12 × 0.3 best × turns rainy	= 0.036 dropped
from Rainy	0.28 × 0.6 best × stays rainy	= 0.168
keep	0.168 route from Rainy wins	larger
walk	× 0.3 Long walk if rainy	= 0.0504
0.0504 · Rainy → Rainy		

Faded rows are dropped; the green bar keeps the score and its route.

Day 3: the same duel, one day later

Starting from the Day 2 winners: Sunny 0.0896 and Rainy 0.0504

Best way into Sunny, Day 3		
from Sunny	0.0896 × 0.7 best × stays sunny	= 0.06272
from Rainy	0.0504 × 0.4 best × turns sunny	= 0.02016 dropped
keep	0.06272 route from Sunny wins	larger
walk	× 0.2 Short walk if sunny	= 0.012544

0.012544 · Rainy → Sunny → Sunny

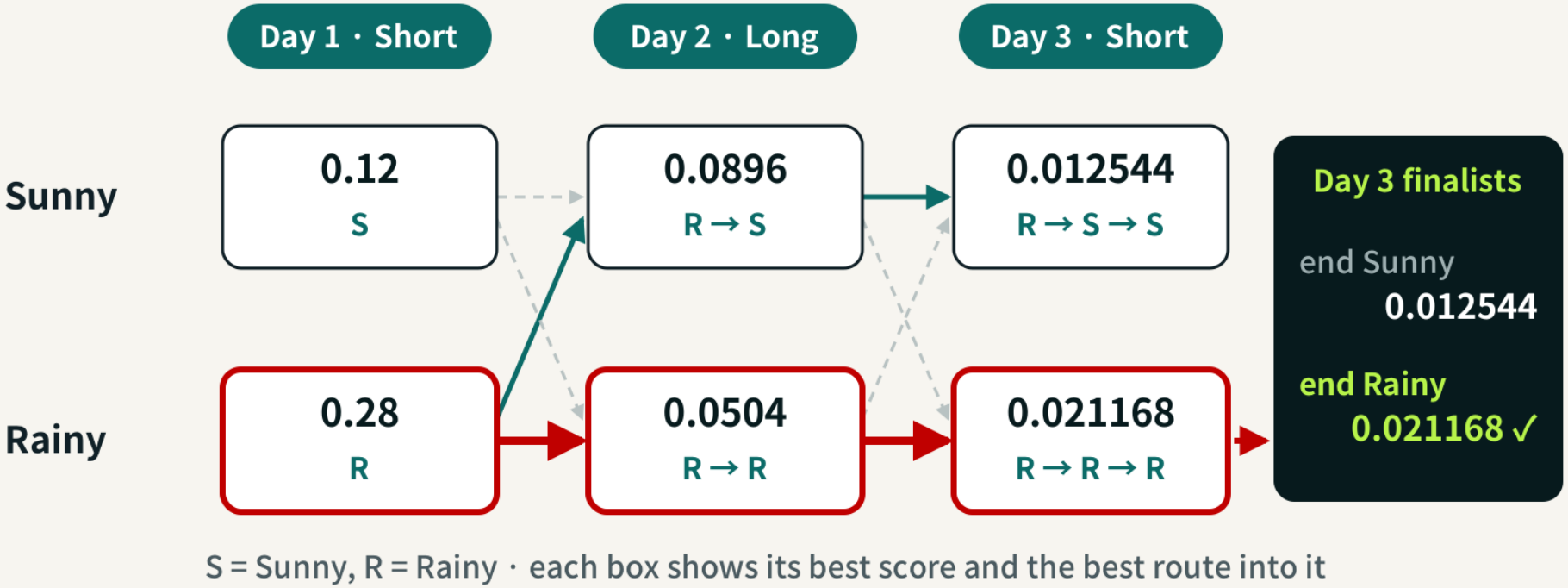
Best way into Rainy, Day 3		
from Sunny	0.0896 × 0.3 best × turns rainy	= 0.02688 dropped
from Rainy	0.0504 × 0.6 best × stays rainy	= 0.03024
keep	0.03024 route from Rainy wins	larger
walk	× 0.7 Short walk if rainy	= 0.021168

0.021168 · Rainy → Rainy → Rainy

Two finalists remain: Sunny 0.012544 and Rainy 0.021168.

The best story ends Rainy, and its route is stored in the winning box

Compare the two Day 3 finalists, then read the route written in the winner



S = Sunny, R = Rainy · each box shows its best score and the best route into it

※ 0.021168 is also the largest of the eight story chances from the slow method

Day 2 Sunny scores higher, yet the best story passes through Day 2 Rainy

Both candidates end Rainy on Day 3; the overnight step decides between them

Through Sunny on Day 2

Day 2 best 0.0896 × 0.3 sunny turns rainy × 0.7 Short walk if rainy = 0.018816 loses

Through Rainy on Day 2

Day 2 best 0.0504 × 0.6 rain stays rain × 0.7 Short walk if rainy = 0.021168 winner ✓

※ From Sunny the weather must switch to rain (0.3); from Rainy it just stays (0.6)

Forward measures how likely; Viterbi finds which story

Same picture, same Day 1 numbers, one word changed

	Forward	Viterbi
Question	**How likely is the walk log?**	**Which weather story is most likely?**
At each box	**Add both incoming routes**	**Keep the larger route**
Remembers	**One running total**	**Best score and its route**
Answer here	**0.08548**	**Rainy, Rainy, Rainy (0.021168)**

Both work day by day, so the effort grows slowly with the number of days.